

SWARNA VAIDEHI MAKHANA HUB

5-Year Scaling Roadmap, Business Model & Bank Loan DPR

Prepared Under the Strategic Framework of:

ICAR-National Research Centre for Makhana, Darbhanga
GI-Tagged Agricultural Crop Expansion Initiative (Bihar, India)

Initial Operational Phase: Outsourced/Decentralized Manual Popping Clusters

Secondary Operational Phase: 5-Ton Daily Capacity Fully Mechanized Processing Line

Date of Issue: May 2026

Classification: strictly confidential / bank loan proposal evaluation

1. Scientific Cultivation Protocol & Value-Add Guide

The elite variety '**Swarna Vaidehi**' (first released on 15.11.2013 and notified on 30.07.2022 by CVRC) developed by the ICAR-NRC for Makhana, Darbhanga represents a massive leap forward. Optimized for field-based agricultural ecosystems (as opposed to natural wetland ponds), this variety ensures high yield stability, standard sizing, and safe manual farming operations.

Step-by-Step Cultivation Protocol:

- **Nursery Bed Sowing (Nov-March):** Raw seeds broadcasted at a low seed rate of **20 kg/ha** (ponds require 90 kg/ha) in clayey mud nursery beds with an area of 500 m²/ha. Seedlings are ready at 75 days old in February.
- **Field Transplanting (Feb-March):** Healthy seedlings are transplanted into pre-bunded ploughed fields at a standard row spacing of **1.20 m**.
- **Water Depth Control (March-August):** Water is maintained at a shallow depth of strictly **30 - 45 cm**. This saves 80% of water compared to natural deep ponds and guarantees workers can stand during harvest, bypassing the dangerous deep diving associated with pond monoculture.
- **Precision Nutrient Regimen:** Soil nutrition requires NPK applied at **100:60:40 kg/ha** in split doses along with 10 t/ha Farm Yard Manure (FYM). Core secondary micro-nutrients: Zinc (5 kg/ha), Magnesium (5 kg/ha), and Boron (5 kg/ha) combined with foliar spray of 0.1% CuSO₄ solution ensure heavy seed shell development.
- **Integrated Pest Management (IPM):** Sucking insects like Makhana Aphids (*Rhopalosiphum nymphaeae* L.) and Singhara Beetle leaf-perforators are controlled organically using Coccinellid beetles, Metarhizium anisopliae or Beauveria bassiana mycological agents (@ 1x10⁸ cfu/g), or 1% Neem oil. Target chemistry includes Flonicamid 50% WG @ 0.4 g/L or Chlorantraniliprole 18.5 SC @ 0.3 mL/L.
- **Root Borer Grub Control:** Managed via granular Chlorantraniliprole 0.4% GR @ 6 kg/acre applied to standing water. Leaf blight is controlled with Dithane M-45 (0.3%) sprays.
- **Harvesting & Sweeping (July-August):** Mature seeds (berries grow 6-8 cm, plants bear 8-15 fruits containing 20-200 seeds of 0.8-1.5 cm size) collected standing up using Gaanj nets. Yield ranges from **3.0 - 3.5 t/ha**.

Premium Value-Added Food Product Chemistry:

- **Makhana Ragi Cookies (Biscuits):** Micronized Euryale ferox seed perisperm flour and finger millet (ragi), rich in essential amino acids (Glutamine, Arginine, Methionine) and calcium (160mg/100g) with stable low fats (0.1%). Retains **58% Gross Margin**.
- **Hypoallergenic Baby Food (Shishu Aahar):** Hypoallergenic weaning flour blending roasted Makhana perisperm (45%), decorticated green gram (30%), and ragi (25%). High essential amino acids index (88.5-91.6%) and optimized 1.2:1 Ca:P ratio. Retains **62% Gross Margin**.
- **Low-GI Functional Pasta:** Extruded pasta made from 50% cold-gelatinized Euryale starch (high resistant amylose starch) and durum wheat semolina. Highly digestible perisperm starchy core keeping GI below 50. Retains **52% Gross Margin**.

2. Distributed Sourcing & Logistics Pipeline

To capture maximum value-addition and avoid localized middleman manipulation, the Hub structures direct farm-gate buybacks across North Bihar districts (Darbhanga, Madhubani, Purnia):

- 1. Farm-Gate Direct Buyback:** Raw seeds are purchased directly from bunded fields using digital moisture testers. Prices are locked based on seed size grades: ■150/kg for Grade-A (>1.2cm), ■130/kg for Grade-B (0.9-1.2cm), and ■100/kg for Grade-C (<0.9cm).
- 2. Primary Sun-Drying:** Seeds are spread over HDPE sheets in dedicated sun yards to dry, reducing moisture from 28% down to 12-14%. This prevents aflatoxins and boosts popping ratios.
- 3. Rotary Nested Sizing:** Sorting into 5 distinct sizing drums. Uniform grades are critical to guarantee even heat distribution during roasting.
- 4. Decentralized Popping Networks:** Graded seeds are distributed to skilled traditional Mallah families for popping. Families use the 'double-roast and mallet' method, receiving ■35/kg job-work fees. This keeps initial CAPEX at zero while creating rural livelihood opportunities.

3. Financial Detailed Project Report (DPR)

Table 3.1: 1-Hectare Field Cultivation Sheet (Swarna Vaidehi)

Item Description / Input Spec	Cost Category	Cost (INR)
Land Lease of bunded field (1 Year)	Fixed Lease Cost	Rs. 25,000
Field preparation & perimeter leveling	Tractor & Earth Moving	Rs. 15,000
Nursery seedlings & certified seed stock	Seeds (20 kg/ha)	Rs. 8,000
NPK Fertilizer & micro-nutrient nutrition	Soil Inputs (Zn, Mg, B)	Rs. 12,000
Integrated Pest Management bio-agents	Crop Protection	Rs. 5,000
Transplanting, weeding & sweeping labor	75 Mandays @ Rs. 600	Rs. 45,000
Irrigation pump diesel fuel	Water management	Rs. 10,000
TOTAL CULTIVATION COST (OPEX)	Total Operational Expense	Rs. 1,20,000
Raw Seed Revenue (2800 kg @ Rs. 150/kg)	Market Seed Sales	Rs. 4,20,000
NET PROFIT PER HECTARE	Net Cultivator Return	Rs. 3,00,000

Table 3.2: Year-1 Bootstrapped Processing Ledger (30 Tons seeds)

Operational setup with *zero machinery purchase*, utilizing outsourced traditional Mallah popping clusters for double roasting and mallet popping.

Operational Component	Specification & Scale	Cost / Value (INR)
Raw Seed Procurement	30,000 kg seeds @ average Rs. 130/kg	Rs. 39,00,000
Drying & Sorting Labor	150 Mandays @ Rs. 800 at hubs	Rs. 1,20,000
Logistics & Transport	Farm-gate to processing clusters	Rs. 1,50,000
Manual Popping Job-work	11,700 kg popped makhana @ Rs. 35/kg	Rs. 4,09,500
Branding & Pouch Packaging	46,800 custom polymer bags	Rs. 1,20,000
Warehouse Rent & Overheads	12 months general hub costs	Rs. 1,50,000
TOTAL SOURCING & PROCESSING OPEX	Total Sourced Cost	Rs. 48,49,500
Revenue: Premium Popped Grade-A	8,000 kg sold wholesale @ Rs. 550/kg	Rs. 44,00,000
Revenue: Standard Popped Grade-B	3,000 kg sold wholesale @ Rs. 400/kg	Rs. 12,00,000
Revenue: Broken / Phulla Grade-C	700 kg sold for weaning flour @ Rs. 200/kg	Rs. 1,40,000
Revenue: Makhana Shell Waste	18,000 kg sold as mulch @ Rs. 3/kg	Rs. 54,000
TOTAL REVENUE FROM POPPING	Total Gross Returns	Rs. 57,94,000
NET HUB PROCESSING PROFIT (Y1)	Net Processing Hub Profit	Rs. 9,44,500

Consolidated Profit Synthesis: By integrating 1-Hectare self-cultivation (■3,00,000) and the 30-ton hub ledger (■9,44,500), the business achieves an aggregate net profit of ■12,44,500 in Year 1 with zero machinery CAPEX. This fulfills the strategic objective.

Table 3.3: Future 5-Ton Mechanized Scaling Phase (CAPEX Breakdown)

CAPEX values are strictly derived from the ****ICAR-NRC Makhana core food machinery price records**** (File 1):

Machinery CAPEX Items	Technical Details / Hourly Output	Cost (INR)
Secondary Popping Machine	Impact-based popping at 270 °C (25-30 kg/h)	Rs. 8,00,000
Primary Seed Roasting Machine	Roasts raw seeds at 270 °C (100 kg/h)	Rs. 1,40,000
Rotary Nested Seed Grader	Separates raw seeds into 7 grades (180 kg/h)	Rs. 2,50,000
Makhana Seed Washer	Cleans raw seeds via friction (120-180 kg/h)	Rs. 1,60,000
Makhana Seed Grader	Sand-suction slurry sorting (0.5 t/h)	Rs. 1,30,000
Cabinet Seed Dryer	Temp range 30-220 °C (20-30 kg/h)	Rs. 2,00,000
Popped Makhana Grader	Sorts puffs into 7 size classes (100 kg/h)	Rs. 2,00,000
Automatic FFS Packing Machine	Multi-head weigher packing (6-8 t/day)	Rs. 8,00,000
Makhana Polishing Unit	Polishes finished pops to standard textures	Rs. 1,00,000
Warehouse Civil Works & Drying Slab	Concrete slab and dust-free sorting floor	Rs. 15,00,000

Solar Rooftop Micro-Grid	15kW solar array with 10kW battery backup panels	Rs. 5,00,000
TOTAL CAPEX REQUIREMENT	Automated Infrastructure CAPEX	Rs. 48,50,000

4. Integrated Farming System (IFS) & Excellence Analysis

To maximize field efficiency, the traditional makhana monoculture is compared against an ****Integrated Farming System (IFS)****. Stocking Indian Major Carps (Catla, Rohu, Mrigal) and planting a winter crop of Water Chestnut (Singhara) in the same field body increases resource efficiency.

Operational Metric	Traditional Field Monoculture	Makhana + Carps + Water Chestnut IFS
Water Depth Maintenance	30 - 45 cm (Shallow)	50 - 60 cm (For carp movement)
Primary Makhana Yield	2,800 kg/ha	2,600 kg/ha (Due to fish channels)
Carps Fish Yield	0 kg (None)	1,200 kg/ha (Synergistic carp growth)
Water Chestnut Yield	0 kg (None)	3,000 kg/ha (Secondary winter crop)
Total Operational Cost	Rs. 1,20,000	Rs. 1,80,000 (Includes fingerlings/seeds)
Total Gross Revenue	Rs. 4,20,000	Rs. 7,30,000 (Makhana + Fish + Chestnut)
NET FARM PROFIT PER HA	Rs. 3,00,000	Rs. 5,50,000 (An 83% Profit Increase)

Organic Nutrient & Pest Protection Sourcing:

- **Azolla Pinata:** Grown floating in early makhana vegetative phases, this bio-fertilizer fixes nitrogen, reducing urea dependency by **40%**.
- **Neem Cake (250 kg/ha):** Basal dressing during soil prep, naturally suppressing root-knot nematodes and eliminating synthetic chemistry.
- **Garlic-Chilli botanical sprays:** Made locally, protecting the early foliage against case-worms (*Nymphula crisonalis*) and aphids without chemical residues.
- **Trichoderma Viride:** Fungal treatment of nursery seed stock to prevent damping-off.

Global Export Positioning:

- **United States (US):** Organic puffed snacking segment, fetching an ****85% price premium****.
- **European Union (EU):** Rigid multi-residue pesticide limits and high aflatoxin compliance (< 2 ppb), yielding a ****110% price premium****.
- **Bihar Bazar Integration:** Live integration with `bihar-bazaar.vercel.app` enables voice/SMS price discovery, bypassing local middlemen and providing transparent batch tracking for international buyers.